

Zhi Chen

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EDUCATION

- **Stony Brook University** Stony Brook, NY
Doctor of Philosophy - Physics; *Aug 2023 - Present*
- **Stony Brook University** Stony Brook, NY
Master of Arts - Physics; GPA: 3.94/4.00 *Aug 2021 - May 2023*
- **Stony Brook University** Stony Brook, NY
Bachelor of Science - Physics; GPA: 3.96/4.00; Summa Cum Laude *Aug 2017 - May 2021*

RESEARCH EXPERIENCE

- **PhD Thesis Research/ Research Assistant** *Aug 2023 - Present*
Graduate Student Research
- **Master Thesis Research / Research Assistant** *Jan 2022 - Aug 2023*
Graduate Student Research
 - Primary Project Description: Investigated how different reaction networks, screening methods, and integration methods affect thermonuclear flame propagation for X-Ray Bursts using *Castro*, an adaptive mesh, astrophysical radiation hydrodynamics simulation code.
 - Astronomy Software Development: Worked collaboratively in a team to develop astronomy software called *pyro2*, *Pynucastro*, *Microphysics*, and *Castro*.
- **SULI Program at Brookhaven National Lab + Continued Research** *Jun 2021 - Jan 2022*
Summer Internship + Student Research
 - General Setting: Worked in a small group developing a technique called two-photon interferometry, which has the potential of measuring the relative separation of two luminous objects with high precision.
 - Data Analysis using Python: Analyzed laboratory data by making and fitting various plots using Python.
 - Two-Photon Interferometry Simulation using Python: Developed a *simulation code* that simulated the theoretical observational signal through two-photon interferometry given a set of telescopes and star pairs using Python.
 - Additional Features: Data Sampling and MCMC: Added features such as data sampling and Markov Chain Monte Carlo algorithm to predict the precision of the relative separation between the star pair.
- **SULI Program at Brookhaven National Lab** *Jun 2020 - Aug 2020*
Summer Internship
 - General Setting: Worked in an Astro-group for building a radio-telescope to detect 21-cm emission from hydrogen
 - Primary Role: Developed additional features for *imcurio*, which is a simulation code for 21 cm intensity mapping observations that simulates a fixed set of visibilities for a fixed telescope and fixed sky using Python.

PUBLICATIONS

- Zingale, M., Bhargava, K., Brady, R., **Chen, Z.**, Guichandut, S., Johnson, E., Katz, M. & Clark, A.: The Challenges of Modeling Astrophysical Reacting Flows. doi: 10.1088/1742-6596/2997/1/012007
- Clark, A., Johnson, E., **Chen, Z.**, Eiden, K., Zingale, M., Boyd, B., Johnson, P., & DaCosta, L.. (2024): *pynucastro 2.1: an update on the development of a python library for nuclear astrophysics*. doi: 10.1088/1742-6596/2742/1/012003
- **Chen, Z.**, Johnson, E., Katz, M., Clark, A., Boyd, B., & Zingale, M.. (2024): *A Framework for Exploring Nuclear Physics Sensitivity in Numerical Simulations*. doi: 10.1088/1742-6596/2742/1/012021
- Zingale, M., **Chen, Z.**, Johnson, E., Katz, M., & Clark, A.. (2024): *Strong Coupling of Hydrodynamics and Reactions in Nuclear Statistical Equilibrium for Modeling Convection in Massive Stars*. doi: 10.3847/1538-4357/ad8a66
- Zingale, M., **Chen, Z.**, Rasmussen, M., Polin, A., Katz, M., Clark, A., & Johnson, E.. (2023): *Sensitivity of Simulations of Double Detonation Type Ia Supernova to Integration Methodology*. doi: 10.3847/1538-4357/ad3441
- **Chen, Z.**, Zingale, M., & Eiden, K. (2023): *Sensitivity of He Flames in X-Ray Bursts to Nuclear Physics*. doi: 10.3847/1538-4357/acec72
- Clark, A., Johnson, E., **Chen, Z.**, Eiden, K., Willcox, D., Boyd, B., Cao, L., DeGrendele, C., & Zingale, M.. (2023): *pynucastro: A Python Library for Nuclear Astrophysics*. doi: 10.3847/1538-4357/acbaff
- **Chen, Z.**, Nomerotski, A., Slosar, A., Stankus, P., & Vintskevich, S.. (2022): *Astrometry in two-photon interferometry using Earth rotation fringe scan*. doi: 10.1103/PhysRevD.107.023015
- Keach, M., Bellavia, S., **Chen, Z.**, Crawford, J., Dolzhenko, D., Figueroa, E., ... Vintskevich, S. (2022): *Increasing baselines and precision of optical interferometers using two-photon interference*. A. Mérand, S. Sallum, & J. Sanchez-Bermudez, *Optical and Infrared Interferometry and Imaging VIII*. doi:10.1117/12.2632122

CONFERENCES AND MEETINGS

- Contributed Talk at CeNAM Frontiers in Nuclear Astrophysics Meeting, 2026: *Simulating Lateral Flame Propagation in Type I X-ray Bursts*
- Poster Presentation at OLCF User Meeting, 2025: *Simulating X-ray Burst Flame Propagation: From Parallel Plane to Full Star*
- Poster Presentation at Rise.Time Meeting, 2024: *Adaptive Nuclear Statistical Equilibrium for Type Ia SN*
- Poster Presentation at Astronum Meeting, 2023: *Sensitivity of He Flames in X-ray Bursts to Nuclear Physics*
- Contributed Talk at the main CeNAM Frontiers in Nuclear Astrophysics Meeting, 2023: *Sensitivity of He Flames in X-ray Bursts to Nuclear Physics*
- SULI Internship Presentation at Brookhaven National Laboratory, Summer 2021: *Two-Photon Interferometry Simulations*
- SULI Internship Presentation at Brookhaven National Laboratory, Summer 2020: *Hydrogen Intensity Mapping Simulator: interpolation improvements and radio point source compatibility with application to GaLactic and Extragalactic All-Sky MWA Survey (GLEAM) dataset*

HONORS, GRANTS, AWARDS, AND ACHIEVEMENTS

- Gerry Brown Award (2025): Awarded for excellent research and travel funding
- Lourie Summer Fellowship (2024)
- Di-Tian Prize (2023): Awarded for excellent research and travel funding
- Stony Brook University, Academic Dean's List, 2017-2021
- NYS School Academic Excellence Scholarship

SKILLS SUMMARY

- Languages: Python, C++, Bash
- Tools: GIT, Latex, Sphinx, Doxygen, yt
- Platforms: Linux, Web, Windows